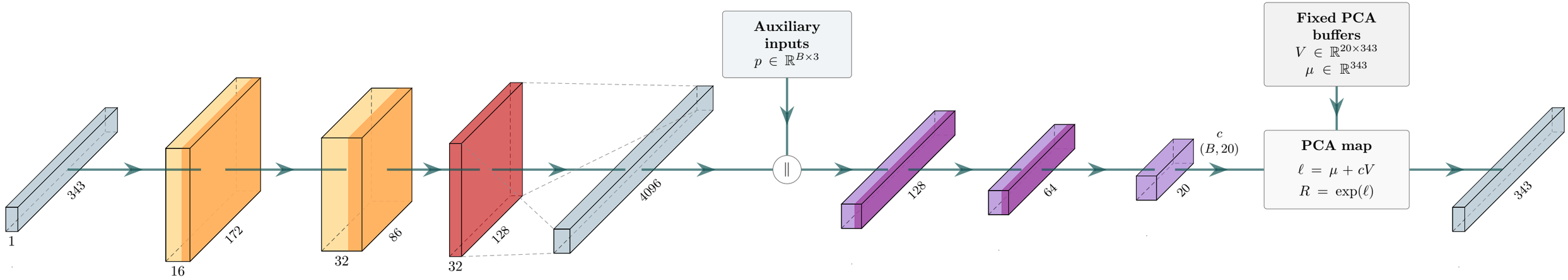
DirectConvPINN

$N = 343$ wavelength bins $P = 3$ auxiliary inputs $K = 20$ PCA components $H = 128$

`forward(x, p)`: **coefficient prediction**

`reconstruct(c)`: **fixed PCA map**



Normalized input	**Conv1d + GELU**	**Conv1d + GELU**	**Adaptive avg. pool**	**Flatten**	**Concatenate**	**Linear + GELU**	**Linear + GELU**	**Linear**	**Non-trainable**	**Calibration ratio**
x_{norm}	$k = 7,\ s = 2,\ p = 3$	$k = 5,\ s = 2,\ p = 2$	$L_{\text{out}} = 128$	$32 \times 128 = 4096$	$4096 + 3 = 4099$	$4099 \to 128$	$128 \to 64$	$64 \to 20$	PCA fitted on training stars	$R \in \mathbb{R}^{B\times 343}_{>0}$

Input normalization ($x \in \mathbb{R}^{B\times N}$)

$s = \max_\lambda |x| + 10^{-20}, \qquad x_{\text{norm}} = \text{unsqueeze}(x/s)$

Tensor labels (batch axis B omitted)

Feature maps: channels and wavelength bins; vectors: features.
Block dimensions are schematic; pooling produces 128 bins.

Output semantics

Final linear layer: $W = 0$, $b = 0$ initially, hence $R = \exp(\mu)$.
Calibrated flux is computed outside the class: $\widehat{Y} = x \odot R$.

Auxiliary inputs: BP−RP, G, RUWE; standardized before the model.

Legend Conv1d · Adaptive average pooling · Linear · Bands: GELU